MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON, LOOP

1. SCOPE

1.1 Scope.- This specification covers one type of nylon loop tape used as a lacing tape in flight garments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods

Military

MIL-P-6065	Packaging, Packing, and Marking of Cloth, Fabric, and Webbing

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material

3.1.1 Yarn.- Unless otherwise specified, the nylon yarn used in the tape shall be made from 210 denier, bright 34 filament, high tenacity, polyamide prepared from hexamethylene diamine and adipic acid or its derivatives. It shall have a melting point of 482°F ± 10°F when tested as specified in 4.3.2.1.

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3.1.2 Loop cord.- The loop cord used in the manufacture of the lacing loop tape shall be constructed in such a manner and with such yarn tension that the core yarns are held firmly in place to prevent puckering of the core yarns when released from stress. This cord shall conform to the requirements of table 1 when tested as specified in 4.3.

TABLE I Cord Requirements

	Sleeve	Core	Cord
Yarn size, denier	210	210	-
Yarn ply (min)	3	2 or 3	-
Yarn twist, turns per inch (min)			
spin	8S	8S	-
ply	6Z	6Z	-
Number of carriers	20	-	-
Ends per carrier	1	-	-
Number of ends	20	6 or 9	-
Picks per inch	26-28	-	-
Weave	plain(1 up 1 down)	-	
Thickness, inch ± 0.010	-	-	0.075
Breaking strength, pounds (min)	-	-	150
Yards per pound (min)	-	-	200

3.2 Tape construction and physical properties.- The finished tape shall conform to the requirements specified in table II and the following subparagraphs when tested in accordance with 4.3.

TABLE II Construction and Physical Properties

Yarn ply	
warp	Singles
filling	3
Thread count, min	
Face warp, full width	38
Back warp, full width	37
Cord warp, full width	1
Total warp, full width	76
Filling, picks per inch	60
Thickness, inches (exclusive of cord)	0.030-0.040
Weight, yards/pound (max)	67
Width, Inches ± 1/32	
Tape and cord woven together	5/16
Tape and cord loop	7/16
Breaking strength, pounds (min)	375

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3.2.1 Weave.- The finished loop tape shall be woven in accordance with the weave diagram shown in figure 1.

3.2.2 Length relationship between loop cord and finished tape.- The finished tape shall contain a length of 45 ± 1 inch of cord for each yard of tape in order to produce the desired loop opening.

3.3 Color.- The color of the tape shall match Air Force sage green shade no. 520 (see 6.3).

3.3.1 Colorfastness for sage green 520 tape.- The finished tape shall have a colorfastness rating of "good" to light, laundering, crocking and perspiration when tested as specified in 4.3.2.2.

3.4 Finish

3.4.1 Nonfibrous material.- The finished tape shall not have more than 4.25 percent nonfibrous material when tested as specified in 4.3.

3.4.2 Acidity.- The finished tape shall have a pH value of 5 to 9 when tested as specified in 4.3.

3.5 Length of spool.- Unless otherwise specified, each spool of tape shall contain approximately 50 yards. No spool shall contain more than 2 pieces, and no piece shall be less than 10 yards in length.

3.6 Identification of product.- Each spool of tape shall be marked for identification in accordance with Specification MIL-P-6065.

3.7 Workmanship.- The finished tape shall be clean, evenly woven and shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of the webbing shall be classified as acceptance tests.

4.2 Acceptance tests.- Acceptance tests shall consist of:

a. Sampling plan A and tests
b. Sampling plan B and tests

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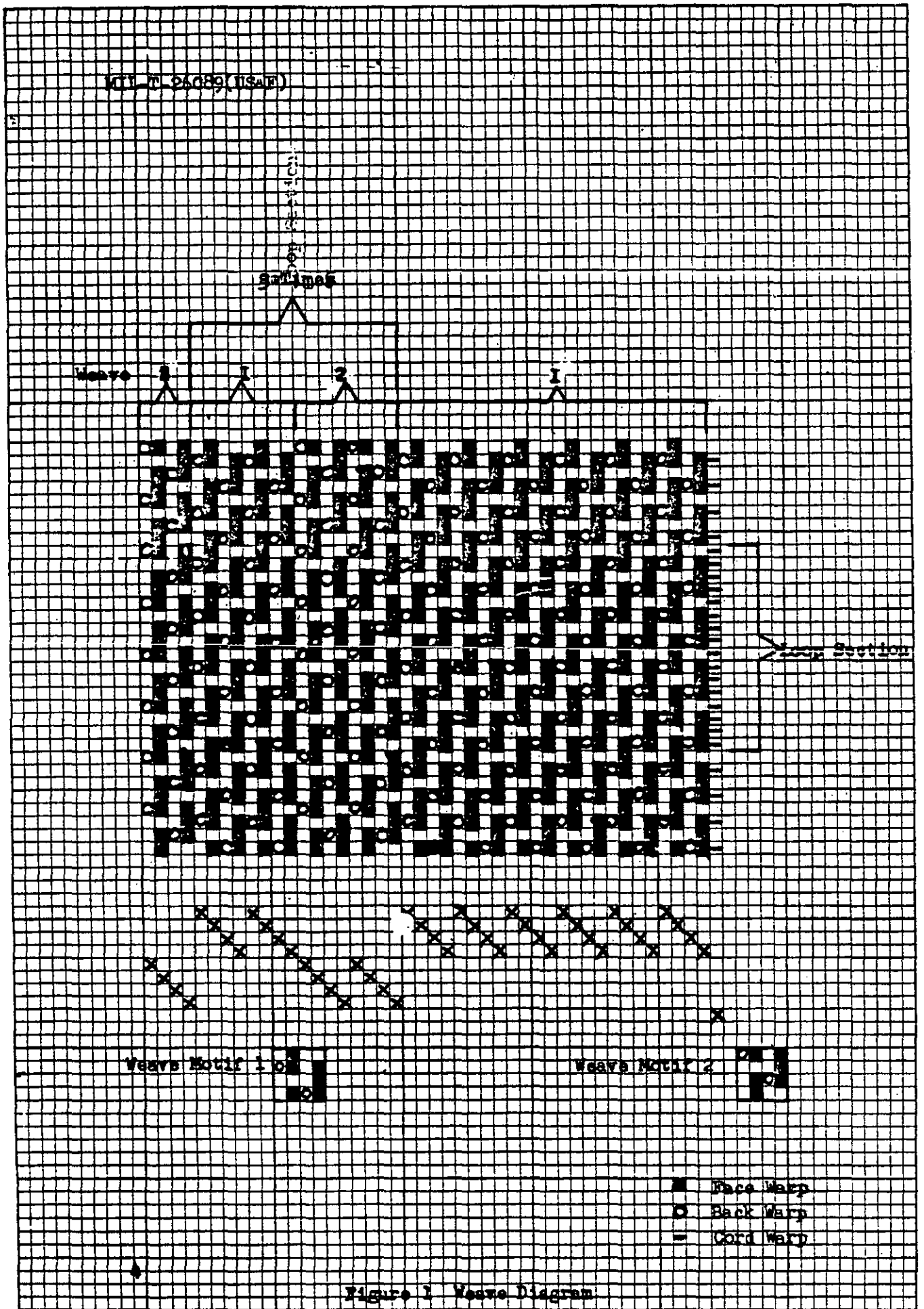


Figure 1. Weave Diagram

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4.2.1 Sampling plan A.- For the inspection of the product tests, sampling shall be conducted in accordance with MIL-STD-105. Samples selected shall be subjected to the tests specified in 4.3.1 and subparagraphs thereto.

4.2.1.1 MIL-STD-105.- Except where otherwise indicated, inspection shall be in accordance with the provisions set forth in MIL-STD-105. When MIL-STD-105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.2.1.2 Acceptance quality levels.- Acceptance quality levels shall be as follows:

a. For yard-by-yard examination (4.3.1.1) 2 major defects and 15 total defects per 100 units (yards).

b. For overall examination (4.3.1.2) 1 defect per 100 units (pieces).

c. For examination for length of spools (4.3.1.3) - 1 defect per 100 units (spools).

4.2.1.3 Inspection levels.- The inspection levels shall be as follows:

a. Yard-by-yard examination (4.3.1.1).- Level III of MIL-STD-105. The lot size shall be expressed in units of 1 yard each. An approximately equal number of yards shall be examined from each piece in the sample. The number of pieces to be drawn shall be computed as follows:

$$\text{Number drawn} = \frac{\text{Yards in sample}}{35}$$

b. Overall examination (4.3.1.2).- The pieces examined shall be those selected for yard-by-yard examination.

c. Examination for length of spool (4.3.1.3).- The spools examined shall be those spools from which the pieces for yard-by-yard examination were taken.

4.2.2 Sampling plan B.- For the physical acceptance tests, a sample or samples, each at least 15 yards long, shall be taken from each 5,000 yards, or fraction thereof, of finished tape and of the component cord, submitted for acceptance at one time on any contract or order. Each sample shall be suitably marked for identification and shall be subjected to all tests specified in 4.3.2 and subparagraphs thereto.

4.3 Test Methods

4.3.1 Examination of product.- Defects found during the examination shall be classified in accordance with 4.3.1.1, 4.3.1.2, and 4.3.1.3. The Government reserves the right to require examination for any defect not included herein to determine compliance with the requirements of this specification and to classify such defects in accordance with the definitions contained in MIL-STD-105 (see 4.2.1.1).

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4.3.1.1 Yard-by-yard examination.- The required yardage of each piece shall be inspected and visual defects classified as listed in table III. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the tape, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one linear yard.

TABLE III

Classification of Defects

DEFECT	DEFINITION	MAJOR	MINOR
Abrasion mark	Resulting in rupture of individual yarns or plies, distortion in the orientation of threads, dimensional distortion, areas noticeably thinner than adjoining unaffected areas, or in nap sufficient to obscure the identity of any yarns.	X	
Broken or missing end	Any broken or missing end.	X	
Broken or missing pick	Two or more contiguous, regardless of extent.	X	
	Single, more than 1/2 inch the width of the tape.		X
Coarse filling bar, heavy place	Resulting in noticeably increased stiffness or thickness of tape and extending for more than 1/4 inch in the length direction.	X	
	Resulting in noticeably increased stiffness or thickness of tape and extending for 1/4 inch or less in the length direction.		X
Crease or wrinkle (edge folded or rolled)	Hard, embedded.		X
Cut, hole, or tear	Any cut, hole, or tear.	X	

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TABLE III (con't)

DEFECT	DEFINITION	MAJOR	MINOR
Drop-ply	Two or more drop-ply ends in the same length.	X	
	Single drop-ply end (see "fine or light filling bar").		X
Edge cut, torn, or frayed	Any cut, torn, or frayed edge.	X	
Edge beaded or corded	Noticeable increase in edge thickness or misformed edge.		X
Edge loopy	Forming clearly noticeable filling loops or edges tied loosely to body of tape for 2 linear inches or more.	X	
Edge loose (slack)	Resulting in waviness, distortion in orientation of filling, or looseness along edge.	X	
Edge tight	Resulting in noticeable tension along edge, or pucker, waviness, bagginess, or slackness that cannot be flattened by manual pressure.	X	
Scalloped edge	Any noticeable indentation of edge.		X
Fine or light filling bar, light place.	Clearly noticeable.	X	
Floats or skips	Any multiple floats or skips.	X	
	Single float or skip more than 1/4 inch in length in warp or filling direction.	X	
	Single, floating for 1/4 inch or less in warp direction, or up to 1/4 inch, or 1/3 of the width of the tape (whichever is the lesser) in the filling direction.		X

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TABLE III (con't)

DEFECT	DEFINITION	MAJOR	MINOR
Hitch-back crack, open place	Clearly noticeable opening between adjoining picks or warpwise tension area over part of the width, resulting in noticeable light and heavy places.		X
Jerked-in filling, slough-off, slug	More than twice the thickness of the normal yarn.		X
Kinks	More than 3 in any 9 linear inches.	X	
Knots	More than 1 knot in any 9 linear inches.	X	
	Single knot with untrimmed ends extending from surface of tape.		X
Mispick, double pick	Two or more contiguous.	X	
	Single.		X
Slack end	Two or more in same length, jerked in between picks, or forming clearly noticeable loops on surface of tape.	X	
	Single, jerked in between picks, or forming clearly noticeable loops on surface of tape.		X
Smash	Any smash.	X	
Spot, stain, or streak	Any uncleared dirt, rust, grease or oil spot, stain or streak.		X
Tight end	Clearly noticeable.		X
Tight pick or tight filling.	Resulting in rolling of tape (see also "edge scalloped").	X	
Weak, soft, or tender spot	Any weak, soft, or tender spot.	X	
Width	More than 1/32 inch narrower or wider than specified.		X
Wrong draw	Extending for more than 9 inches.		X

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4.3.1.2 Overall examination.- The unit of product for this examination shall be one piece. The tape shall be inspected for the following defects:

	Defect
Finish	Objectionable odor
	Unclean throughout
Shade	Off shade, i.e., not within established tolerances
	Shaded, cloudy, or streaky color throughout
	Poor penetration, two sidedness
Weaving	Uneven weave throughout

4.3.1.3 Examination for length of spool.- The spool shall be examined for gross length. Any gross length found to be less than the specified minimum length or any gross length found to be more than 2 yards below the gross length marked on the piece ticket shall be considered as a defect with respect to length. The unit of product for this examination shall be one spool.

4.3.2 Physical acceptance tests.- The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table IV, 4.3.2.1 and 4.3.2.2 shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. Failure of any sample of tape to conform to the requirements specified in section 3 shall be cause for rejection of the lot represented.

TABLE IV.- Test Methods

Test	Method No.
Nonfibrous materials 1/	2611
Acidity	2811
Breaking strength	4102
Width	5020
Thickness	5030
Twist	4054
Weight	5040
Thread count	5050
Colorfastness to laundering	5614
Colorfastness to crocking	5650
Colorfastness to light 2/	5660
Colorfastness to perspiration	5682

1/ The paragraphs on water soluble material, weight of desized specimen, residual ash, and total inorganic material are not applicable.

2/ The time of exposure shall be 20 standard fading hours.

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4.3.2.1 Nylon melting point.- A suitable device, so constructed as to assure accurate determination of the point at which melting occurs, shall be used to determine the melting point of the nylon. The temperature of the device containing the nylon yarn may be raised rapidly to 400°F and then gradually raised until the nylon begins to liquefy. This temperature shall be considered the melting point, and the average of ten specimens shall be reported.

4.3.2.2 Color matching.- Dyed tape shall be matched to the approved standard shade under natural (north sky) daylight, or artificial daylight having a color temperature of 7500°K. and shall be a close approximation of the standard shade under incandescent lamplight at 2800°K.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.- Shipments shall be packaged and packed in accordance with MIL-P-6065.

5.3 Marking of shipment.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be: "Tape, Textile, Nylon, Loop."

6. NOTES

6.1 Intended use.- The tape covered by this specification is intended for use in flight clothing.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Quantity desired
c. Provisions for deduction of defects (if any)

6.3 Color standard sample.- The manufacturershould communicate with the procuring activity for access to color standard.

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